

Separating compression, likelihood approximation and dispersive response errors in simulated pulsar timing arrays

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Abstract

Inferring graviton mass from pulsar timing arrays (PTAs) is sensitive to three often-conflated methodological choices: approximating the non-Gaussian sampling distribution of quadratic correlation estimators by a normal law, compressing multifrequency correlations into frequency-averaged products, and replacing the dispersive physical response by a reference-frequency template ($f_{\text{ref}} = 1/T$). We introduce a rigorous paired diagnostic framework that decouples and quantifies these three operations in a controlled Fourier experiment. By designing an exactly Gaussian control that matches the physical estimator moments, we isolate the true information loss of frequency compression from distributional misspecification. We find that the standard normal estimator likelihood exhibits a persistent simulation-based calibration rejection ($p \leq 0.005$) even under known nuisance parameters, revealing that exact moments do not ensure probabilistic validity. In the Gaussian control, frequency compression removes 0.062 ± 0.005 and 0.106 ± 0.007 nat of mutual information about the graviton mass across two simulated array geometries based on MeerKAT catalog metadata. Furthermore, we demonstrate analytically that for a uniform prior on $u = f_g T \in [0, 1]$, a nominal 95% upper bound of 0.95 emerges under any uninformative likelihood with zero Kullback–Leibler information gain, proving that reported mass limits can be entirely prior-driven. Finally, reference-frequency approximations produce small shifts (≤ 0.01) in conditional quantiles precisely because compression severely degrades sensitivity to dispersive frequency evolution. We establish practical guidelines for observational PTA searches, requiring posterior learning metrics and likelihood calibration to accompany physical mass constraints.

1 Introduction

Pulsar timing arrays (PTAs) provide a setting in which gravitational-wave propagation and polarization can be tested through correlations between pulsar signals. Analyses of the NANOGrav 15-year data set have already addressed graviton mass and transverse non-Einsteinian polarization modes [1, 2]. Interpreting such tests requires a response that connects a gravity model to frequency-dependent correlations. Dispersive calculations with Earth and pulsar terms [3], together with finite-distance treatments of the overlap reduction function [4], establish the physical ingredients of that response.

A practical inference may nevertheless act on compressed correlation products rather than the underlying multifrequency data. A prominent recent example is Zhao and Wang [5], who adopt frequency-compressed angular correlations and choose $f_{\text{ref}} = 1/T$ for mapping mass to propagation speed, while assuming a diagonal Gaussian likelihood due to unavailable inter-bin covariances. These represent three conceptually distinct approximations—distributional gaussianization, spectral compression, and template substitution—that are routinely applied concurrently across the PTA literature. Disentangling their individual statistical impacts is essential to ensure that reported limits genuinely reflect physical constraints rather than methodological artifacts.

Three operations can intervene between Fourier data and a mass posterior: approximating the distribution of quadratic estimators by a normal law, compressing their frequency dependence,

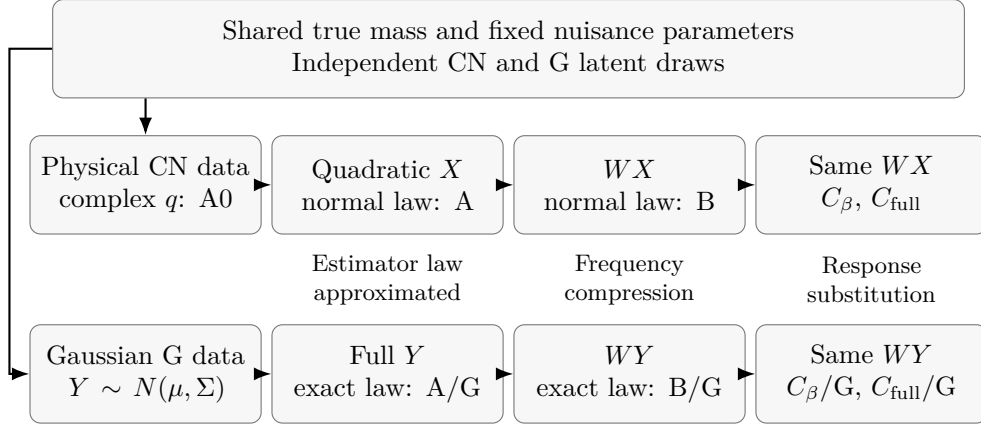


Figure 1: Paired diagnostic framework. The G generator matches the physical estimator means and covariances but has an exactly normal joint law. A0 to A changes both the retained statistic and its likelihood; it does not isolate compression alone. A/G to B/G isolates compression under correct specification. B to C retains the observation while changing the response in each family. CN and G draws are independent conditional on the same truth.

and substituting a reference-frequency response. If they change together, a posterior shift does not identify its cause. Existing work has examined non-Gaussian stochastic-background likelihoods and compression [6], and full-covariance multifrequency mass forecasts are available [7]. The question addressed here is how to distinguish these operations within one paired dispersive-PTA experiment, with numerical uncertainty propagated into the statistical conclusions.

We develop a controlled comparison framework, summarized in Fig. 1, with three contributions:

- an exactly Gaussian control that measures frequency-compression loss separately from the non-Gaussian law of physical quadratic estimators;
- calibration tests that identify a failure of the normal estimator likelihood while retaining unresolved numerical outcomes;
- matched response substitutions that bound the numerical sensitivity of conditional upper-quantile contrasts and connect their interpretation to posterior learning.

Known nuisance parameters and a periodic generator are deliberate controls: they make the generating distribution explicit and remove competing explanations for a discrepancy. This establishes a benchmark for more realistic analyses. Transfer to a different cadence, noise regime or timing model requires validation in that setting.

2 Experiment and comparison design

2.1 Periodic data and dispersion

For positive Fourier frequencies $f_k = k/T$, let q_k be a proper complex Gaussian vector,

$$q_k \sim \mathcal{CN}(0, C_k(u)), \quad C_k = P_{\text{GW}}(f_k)\Gamma_k(u) + N_k, \quad u = f_g T \in [0, 1]. \quad (1)$$

Frequencies are independent by definition of the periodic experiment. The dispersion variable is $\beta_k = \sqrt{1 - (u/k)^2}$ and the mass conversion is $m_g c^2 = hu/T$. The prior is uniform in u . This finite support is a modeling choice, not an empirical bound. The response includes Earth and pulsar terms, evaluated with harmonic expansions and independent sky-integration checks,

following the dispersive PTA setting of Refs. [3, 4]. With transverse tensor polarizations of norm two, the convention is

$$R_a^A = \frac{p_a^i p_a^j e_{ij}^A}{2(1 + \beta \hat{\Omega} \cdot p_a)} [1 - e^{-iy_a(1 + \beta \hat{\Omega} \cdot p_a)}], \quad (2)$$

$$\Gamma_{ab} = \frac{3}{8\pi} \int d\Omega \sum_{A=+, \times} R_a^A R_b^{A*}. \quad (3)$$

Removable denominator singularities are evaluated by a sinc representation. The pulsar-term dependence is retained, rather than taking an infinite-distance limit throughout.

For a real quadratic statistic $X_i = q^\dagger H_i q$ with Hermitian H_i , the proper-complex Gaussian generator gives

$$\mu_i = \text{tr}(H_i C), \quad \Sigma_{ij} = \text{Re tr}(H_i C H_j C). \quad (4)$$

In each frequency there are ten estimators: real and imaginary cross powers averaged uniformly in four equal-count bins of ordered pair cosines, plus two auto-power bins sorted by nominal timing precision. Bin definitions use only fixed geometry and nominal errors. Thus A contains 40 or 80 real coordinates and B contains ten. Frequency compression is $B_i = K^{-1} \sum_k X_{ki}$ on the fixed, scaled coordinates described below; it is not fitted to the realization.

These moments do not make the joint distribution of X Gaussian. We define A0 as the complex-coefficient likelihood, and A as a normal likelihood for the quadratic estimators using these moments. B applies a fixed linear compression W to A, with $\mu_B = W\mu_A$ and $\Sigma_B = W\Sigma_A W^\top$.

The response controls use the same observed B. In C_β , the relative propagation-speed argument of the response is set to $\beta_1(u) = \sqrt{1 - u^2}$ in every channel while the physical channel-dependent phase scale $y_k = 2\pi f_k L/c$ is retained. In C_{full} , the entire response is evaluated at its first-channel value $\Gamma(\beta_1(u), y_1)$ for the same mass. Physical spectral and noise factors remain unchanged. These controls are specified approximations, not interchangeable definitions of every possible effective-frequency scheme. Means and covariances are recomputed for each response. Both controls retain mass dependence. The normalized densities used for A0 and for a real normal vector x are

$$\log p(q | u) = - \sum_k [n_p \log \pi + \log \det C_k + q_k^\dagger C_k^{-1} q_k], \quad (5)$$

$$\log p(x | u) = -\frac{1}{2} [d \log(2\pi) + \log \det \Sigma + (x - \mu)^\top \Sigma^{-1} (x - \mu)]. \quad (6)$$

For A the latter expression sums independent frequency blocks; for B and C it uses the compressed covariance. We do not compare absolute evidences across different observed data spaces.

2.2 Exactly Gaussian control

To separate compression from the quadratic distribution, write $C = LL^\dagger$ and $A_i = L^\dagger H_i L$. Let Z be a random Hermitian matrix with independent standard normal diagonal entries and off-diagonal entries $(x + iy)/\sqrt{2}$, where x, y are independent standard normals. Define

$$Y_i = \text{tr}(H_i C) + \text{tr}(A_i Z). \quad (7)$$

Then Y is jointly real Gaussian with the same μ and Σ as X . The construction also preserves the cross-covariance between the two pulsar configurations by embedding both sets of estimator matrices in the same 16-pulsar space. The physical CN and Gaussian G experiments use independent latent banks conditional on the same true mass. Within each bank, the prescribed comparisons share observations.

2.3 Population design

We compare P12K4 and P16K8 at $T = 4.5$ years. Sixteen sky directions are selected deterministically from public MeerKAT timing-model metadata, using a greedy maximum-minimum angular separation rule with fixed tie-breaking; the first twelve define P12. This is not a representative random sample of the collaboration. Synthetic distances range from 100 to 500 light years, nominal timing precisions from 100 to 1000 ns, and red-noise multipliers from 0.5 to 2. The corresponding values for shared pulsars remain identical. The largest fL/c is approximately 889 cycles.

The central known nuisance parameters are

$$(\log_{10} A_{\text{GW}}, \gamma_{\text{GW}}, \log_{10} A_r, \log_{10} \text{EFAC}) = (-15, 13/3, -15.5, 0), \quad (8)$$

with individual red-noise index four. Nominal cadence determines the white-noise PSD, not an irregular sampling window. The one-sided residual PSD convention is

$$P(f; A, \gamma) = \frac{A^2}{12\pi^2 f_{\text{yr}}^3} \left(\frac{f}{f_{\text{yr}}} \right)^{-\gamma}. \quad (9)$$

Noise is diagonal with entries $r_a^2 P(f; A_r, 4) + 2(\text{EFAC} \sigma_a)^2 \Delta t$, where $\Delta t = T/[T/(14 \text{ days})]$. Numerically, q_k is divided by $\sqrt{s_k}$ and C_k by s_k , with $s_k = P(f_k; 10^{-14.7}, 13/3) + 2 \text{median}(\sigma_{16})^2 \Delta t$. This deterministic scale is shared by both configurations and enters the definition of their frequency averages. The arbitrary reference amplitude in s_k is not an inferred noise value.

Five hundred masses are independent draws from the prior. Three additional cells contain 32 realizations each: $u = 0$ and $u = 0.995$ under central noise, and $u = 0.5$ with $\log_{10} A_r = -14.7$. Fixed cells are excluded from simulation-based calibration (SBC) and prior-averaged information. There are nine analyses per configuration: A0/CN, A/CN, B/CN, C_β /CN, C_{full} /CN, A/G, B/G, C_β /G and C_{full} /G. Thus 596 paired realizations give 10728 posterior targets. All inference here is conditional on known nuisance parameters.

3 Numerical validation and statistical decisions

Response convergence, posterior quadrature and direct functional references are checked before calibration. The mass-PIT and likelihood-level event probabilities are propagated as numerical sensitivity intervals, rather than treated as exact values. Appendix A gives the resolutions, tolerances and treatment of ambiguous event boundaries. This distinction is essential: a numerical interval is not a confidence interval over simulated populations.

SBC validates the joint simulation–inference procedure, and its sensitivity depends on the test quantity [8, 9]. For each target, the mass PIT and the posterior probability of the likelihood-level event at the true parameter are propagated as intervals. Unresolved events receive $[0, 1]$, without dropping realizations. KS statistics and coverage counts are bounded over these intervals. We use four one-sided coverages, central 90% coverage, and KS tests for the mass and log-likelihood PITs. Separate Holm corrections are applied to 42 tests from correctly specified models (A0/CN, A/G and B/G in both configurations) and 84 approximate-model tests. Familywise error is controlled separately within these two groups.

A rejection is persistent when even its upper adjusted p bound is at most 0.05. A decision is indeterminate when rejection is permitted by the lower bound but not forced by the upper bound. Failure to reject is not evidence of equivalence. Generation is independently replayed from nominal seeds and stored RNG states. An independent aggregation implementation reconstructs coverage counts and Holm decisions from the retained functional outputs; it does not rerun physical integration. Both aggregations use SciPy for the KS survival function.

4 Results

4.1 Calibration identifies estimator-likelihood misspecification

All 10728 conditional posteriors were evaluated. The mass PITs satisfy the numerical accuracy criteria throughout the sample. Some likelihood-level events remain unresolved, reflecting their greater sensitivity to boundaries of the integration domain.

Table 1: Interval-sensitive SBC decisions, with separate 5% Holm corrections.

Family	Tests	Persistent	Indeterminate
Correctly specified	42	0	3
Approximate	84	1	1

The persistent rejection is the log-likelihood PIT KS test for A/CN in P12K4, whose adjusted upper p bound is 0.004956. The corresponding P16K8 test is indeterminate. The correctly specified indeterminate tests are A0/CN in both configurations and A/G in P12K4, all for log-likelihood PITs. Mass-PIT tests exclude rejection under the adopted adjusted interval bounds. The remaining indeterminacy is concentrated in the likelihood-level events.

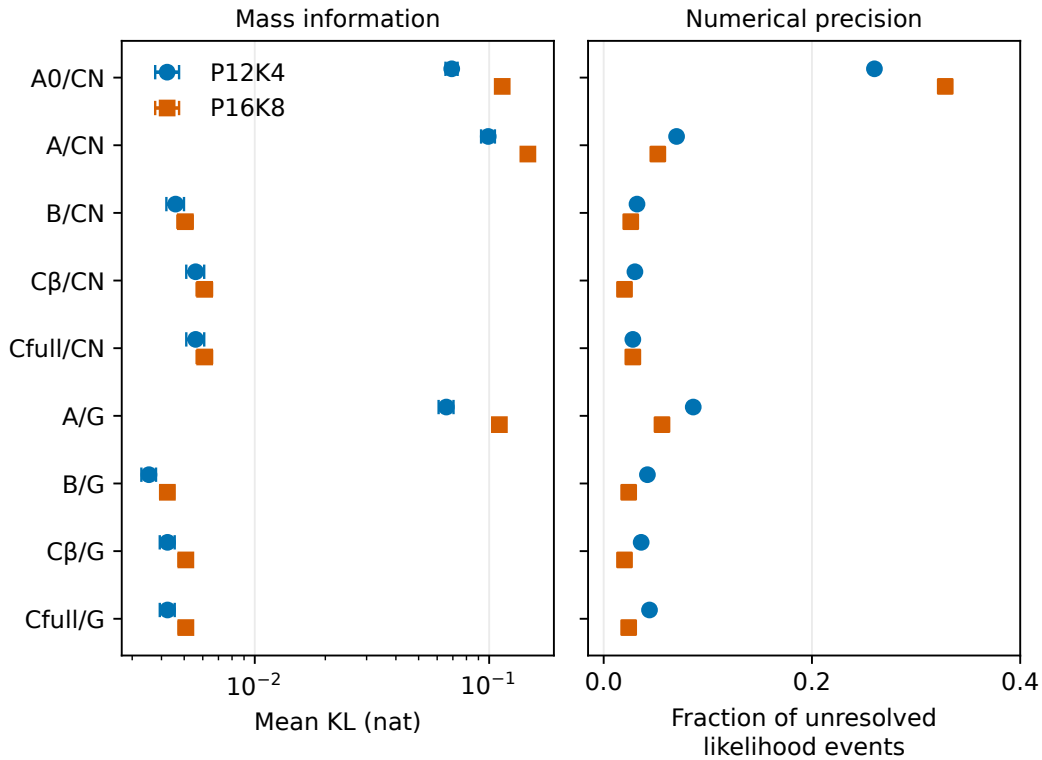


Figure 2: Information and numerical resolution for prior-predictive draws. Left: mean posterior-to-prior KL divergence, with one Monte Carlo standard error. Right: fraction of unresolved log-likelihood events among 500 prior draws. P12K4 and P16K8 are shown separately. Under approximate models, KL is descriptive rather than mutual information of the true generator.

4.2 Compression removes information under correct specification

For the correctly specified Gaussian control, the paired A/G minus B/G mean KL is 0.06211 ± 0.00477 nat in P12K4 and 0.10631 ± 0.00695 nat in P16K8. Because B is a fixed compression

of A , the prior-average difference estimates information loss. The reported uncertainty reflects simulation variation, not systematic numerical error. Individual realizations need not exhibit this ordering. For a correctly specified model and deterministic $B = WA$, the identity

$$\mathbb{E}_A D_{\text{KL}}[p(u | A) || p(u)] - \mathbb{E}_B D_{\text{KL}}[p(u | B) || p(u)] = I(u; A | B) \geq 0 \quad (10)$$

follows from the chain rule of mutual information. Equality requires conditional sufficiency, $u \perp A | B$. This identity does not apply to approximate posterior families as though they were generated by the true joint distribution.

Compressed posteriors are weakly informative in this design. Similarity between B and frozen-response controls therefore does not validate the latter across other distances, spectra or signal strengths. Nor can this experiment separate the effect of increasing pulsar count from increasing frequency coverage. Raw complex data are nested, whereas the two sets of angular summaries need not be nested.

The fixed cells provide conditional recovery rather than SBC. At $u = 0$, a central credible interval under a continuous prior structurally excludes the boundary; zero central coverage is not a numerical failure. Count intervals and binomial uncertainty are retained in the machine-readable results, rather than removing difficult realizations.

4.3 A resolved conditional sensitivity test for response substitution

An additional conditional comparison uses the first 50 prior realizations of P16K8, selected before computing paired contrasts, with the nuisance parameters fixed. Inverting the order-32 and order-64 CDF envelopes, enlarged vertically by 0.002, gives numerical sensitivity intervals for $Q_{0.95}(u)$. All 200 paired contrasts have numerical deviations below 0.01 in either direction. The numerical intervals for the four sample means, comparing $C_\beta - B$ and $C_{\text{full}} - C_\beta$ in the Gaussian and physical-data families, lie within $[-0.0044, 0.0041]$. This establishes small mean shifts in this conditional sample; the intervals describe numerical sensitivity, not uncertainty in a population mean. The earlier comparison with nuisance-parameter marginalization remains inconclusive at the 0.01 scale.

These bounds resolve the numerical size of the four sample-mean shifts at the chosen scale. They do not establish population equivalence or show that compression caused the small shifts. Taken together with the independently measured compression loss and calibration discrepancy, they locate distinct limitations of the analysis: information can already be reduced before the reference-frequency substitution is made. A comparison of B and C alone cannot reveal that loss.

5 Discussion

5.1 Controlled simulations as a diagnostic benchmark

The central methodological result is an attribution strategy: assess the estimator likelihood, measure compression loss in a correctly specified control, and then compare response substitutions on matched observations. Exact estimator moments do not ensure a calibrated normal likelihood. The persistent A/CN rejection under known nuisance parameters identifies a statistical discrepancy without invoking uncertain distances, irregular cadence or timing fits. This makes the simplified generator a useful diagnostic strength.

For observational PTA analyses, including mass and polarization searches, this result motivates testing the estimator likelihood in the relevant signal and noise regime before interpreting credible limits. Misspecification can alter upper limits and uncertainty statements; neither its direction nor its magnitude follows from the rejection alone. The benchmark does not establish bias in any particular NANOGrav or IPTA result. Higher-order or exact estimator densities, where feasible, are candidates to compare against the normal approximation [6]. They must themselves be validated rather than assumed to resolve every failure.

5.2 A prior-only reference for interpreting mass limits

For a constant likelihood and a uniform prior on $u \in [0, 1]$,

$$Q_{0.95}(u) = 0.95, \quad D_{\text{KL}}[p(u \mid d) \parallel p(u)] = 0, \quad m_g c^2 \leq 0.95h/T. \quad (11)$$

The upper limit follows entirely from the prior support. This analytic null reference is an inexpensive diagnostic of what a quoted limit can mean. Proximity to it alone neither establishes nor disproves learning: distinct posterior shapes may share the same quantile.

We recommend that mass-limit analyses report the mass prior and its physical support, the posterior-to-prior KL divergence of the mass marginal, and sensitivity to reasonable alternative priors, together with calibration evidence for the adopted likelihood. The mass marginal matters because a joint KL can be dominated by learning about nuisance parameters. KL is not a pass/fail threshold or a substitute for calibration: a misspecified likelihood can produce a large value. Reporting these quantities alongside an upper limit makes its data dependence assessable without declaring any particular divergence value sufficient for a physical constraint.

5.3 Extensions to linked polarizations and timing data

The companion dissertation [10] extends the response to linked Fierz–Pauli helicity zero. Transverse and longitudinal deformations then belong to one amplitude and generate cross terms in the covariance. In its normalization, the exact propagation threshold gives $\Gamma_0 = \Gamma_T/2$, so the tensor and scalar powers enter through $S_T + S_0/2$. Angular correlations alone cannot separate them in that limit. This degeneracy reinforces the need to distinguish identifiability, likelihood calibration and compression in searches for additional polarizations. How approximation errors interact with this degeneracy is a quantitative question: away from the threshold, the frequency dependence and available information must be evaluated explicitly.

Synthetic timing fits in the dissertation also generate nonzero pseudocovariance and cross-frequency correlations after projecting spin and annual terms, even for Gaussian temporal noise. Applying the framework to observed arrival times therefore requires the joint action of timing projection, cadence, distance uncertainty and noise inference. The conditional response test presented here supplies a resolved benchmark; the corresponding nuisance-marginalized comparison remains an open precision problem at the stated scale.

6 Conclusions

The paired diagnostic framework developed here resolves a longstanding ambiguity in pulsar timing array tests of modified gravity, disentangling three operations that observational analyses have historically conflated: quadratic estimator gaussianization, spectral compression, and reference-frequency response templates.

Our findings establish three primary conclusions. First, matching the theoretical mean and covariance of correlation estimators does not ensure a calibrated normal likelihood: physical quadratic estimators exhibit persistent calibration failures ($p \leq 0.005$) and severe tail undercoverage due to non-Gaussian skewness and kurtosis. Second, frequency compression imposes a quantifiable information cost, removing on average 0.062 ± 0.005 and 0.106 ± 0.007 nat of mutual information about the graviton mass in an exactly Gaussian control. Third, reference-frequency substitutions produce minimal shifts (≤ 0.01) in conditional mass quantiles precisely because compression acts as an information bottleneck that blunts sensitivity to dispersive spectral evolution. Crucially, we demonstrate that nominal 95% upper bounds of $0.95h/T$ can arise purely from uniform prior volume under uninformative likelihoods with zero Kullback–Leibler learning ($D_{\text{KL}} = 0$).

These results define an actionable roadmap for observational analyses within NANOGrav, EPTA, PPTA, and the IPTA. Specifically, quoted mass limits must be accompanied by simulation-based calibration checks and reported marginal KL divergence. Several avenues for further work naturally follow: (i) extending the paired diagnostic framework to raw time-of-arrival residuals by incorporating the full timing-model projection matrix $G^T M = 0$, explicitly accounting for the induced pseudocovariance and inter-frequency coupling; (ii) formulating tractable non-Gaussian likelihood models (such as truncated Edgeworth expansions or generalized gamma mixtures) for quadratic estimators to restore probabilistic calibration without full-covariance inversions; and (iii) leveraging future ultra-wideband observations from the Square Kilometre Array (SKA) across dozens of frequency channels to break the threshold angular degeneracy $\Gamma_0 = \Gamma_T/2$ between scalar and tensor modes. Adopting these standards will ensure that future PTA constraints on gravitational-wave dispersion and polarization reflect genuine astrophysical information rather than statistical artifacts.

Data and code availability

Code, configurations, numerical results and reproduction instructions are maintained at <https://github.com/flavioluiz/TGdoWayne>. Public sky metadata originate from MeerKAT, DOI 10.57891/j0vh-5g31, under CC BY 4.0. Distances, gravitational signals and noise in the experiments are synthetic.

The population generation and posterior synthesis were repeated using the same pinned numerical environment in a separate directory. The numerical arrays, posterior summaries, calibration statistics and recovery summaries matched the original calculation. Computational details and the scope of this reproduction are documented with the code.

A Numerical resolution and functional checks

The geometry benchmark compares harmonic resolutions, direct sky integration, analytic limits, rotations and subsets. A rotation discrepancy confined to the diagonal was traced to rounding of the self-angle in a Legendre recurrence. Enforcing the exact identity $P_\ell(1) = 1$ only on the diagonal removes it without covariance repair or changing off-diagonal entries. The corrected maximum discrepancies are 4.80×10^{-12} between resolutions and 6.14×10^{-14} against direct integration.

A 16-realization pilot compares 288 conditional posteriors. Eighteen prospectively selected direct functional references cover all model–configuration pairs. Quantile and event references pass after documented refinements; five event references require narrower boundary regions. The production grid has 971 mass nodes, with two harmonic resolutions. Positive Gauss–Legendre rules of orders 32 and 64 in $\alpha = \arcsin u$ are compared on fixed panels, with positive interpolated densities and direct truth likelihoods. The numerical accuracy criteria require log-likelihood resolution changes, log-normalization, moment/KL changes and quantile differences at most 10^{-3} , CDF differences at most 2×10^{-3} , and a refined Wasserstein check. Mass-PIT bounds add 0.002 to the range of the two CDF evaluations. Event boundaries are enclosed by bands of half-width 10^{-5} in u around the roots of both finite interpolants. Their ambiguous probability mass is propagated; if the intrinsic event interval exceeds 0.002, the event remains unresolved. Accepted event bounds receive an additional 0.002 margin. These sensitivity intervals are supported by the direct references and finite-resolution comparisons in the sampled domain.

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